Assignment-01: Data Preprocessing for Machine Learning

Deadline: 10th March,2025

Objective: [Mapped with CLO-1 & 2]

Perform data preprocessing on the **Iris dataset** using Python and Scikit-learn. The preprocessing steps will include handling missing values, normalization, feature engineering, and data splitting.

Task 1: Load and Explore the Dataset

1. Load the **Iris dataset** from Scikit-learn.
2. Convert it into a Pandas DataFrame and display the first five rows.
3. Check for missing values and handle them appropriately.

Task 2: Handle Missing Values

1. Introduce some artificial missing values into the dataset.
2. Use **mean imputation** to fill in missing values for numerical features.

Task 3: Feature Scaling

1. Apply **Min-Max Normalization** to scale numerical features between 0 and 1.
2. Apply **Standardization** (z-score normalization) as an alternative.

Task 4: Feature Engineering

1. Create a new feature based on existing ones (e.g., Petal Area = Petal Length × Petal Width).
2. Encode the categorical target variable (species) using **Label Encoding**.

Task 5: Data Splitting

1. Split the dataset into **training (80%)** and **testing (20%)** sets.
2. Print the shape of training and testing datasets.

Bonus Task: Outlier Detection (Optional)

1. Detect outliers using **Z-score** or **IQR (Interquartile Range)**.
2. Remove or replace the outliers based on your chosen method.

Submission Guidelines:

- Submit a Jupyter Notebook (.ipynb) or Python script (.py) with your code and explanations.
- Include comments in your code explaining each preprocessing step.
- Provide a brief summary of how preprocessing improves machine learning models.